

Indirakumar S

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Summary

Backend Engineer-GenAI & Cloud with 2+ years at IBM building production-grade agentic systems, AI-powered automation pipelines, and cloud-native backend services. Specializes in multi-agent orchestration, LLM integration, and end-to-end MLOps with hands-on delivery of systems that handle 10M+ records, automate 1,000+ monthly workflows, and drive measurable operational impact. Proven across the full stack: from LangChain ReAct agents and RAG architectures to Kubernetes deployments, CI/CD pipelines, and distributed data engineering.

Skills

Languages: Python, SQL, Go

Backend & Frameworks: FastAPI, PyTorch, TensorFlow, OAuth2, JWT, REST, gRPC

Cloud & AI Platforms: Azure OpenAI, AWS Bedrock, GCP Vertex AI

Ops: Docker, Kubernetes, Terraform, CI/CD, Openshift, Github Action

GenAI & Agents: LLMs, RAG, LangChain, ReAct Agents, Multi-Agent Systems, Prompt & Context Engineering, NLP, MCP, A2A

Data & Databases: PostgreSQL, Cassandra, Redis, Pinecone, AzureDB

Work Experience

Backend Engineer - Genai/Cloud, IBM – Bangalore

Sep 2023 – Present

- Designed and deployed a GenAI portal for QA teams that auto-generates complete test suites including test cases, documentation, and Postman collections from uploaded Confluence pages, BRDs, DB schemas, and API specs. Engineered and optimized **multi-agent prompts** to interpret system behavior and map API contracts, cutting **1–2 days of manual effort per 20 API contracts down to a single run**. Containerized and deployed the agent stack using Docker for consistent, team-wide hosting.
- Designed and deployed **production-grade Generative AI systems and backend platforms**, focusing on multi-agent architectures, LLM integration, and cloud-native scalability. Built a **multi-agent GenAI framework** for legacy application modernization, enabling automated codebase analysis, business logic extraction, and generation of cloud-native microservices—significantly reducing manual effort and accelerating migration timelines.
- Developed an **AI-powered ticket resolution and audit system** using LangChain-based ReAct agents, capable of analyzing support tickets, generating resolution suggestions, and automating quality audits for **1,000+ tickets per month**, contributing to **\$1M+ in operational savings** through reduced resolution time and improved efficiency.
- Designed a **RAG-based enterprise knowledge system** using Azure OpenAI and Cognitive Search to enable real-time retrieval of internal documentation, eliminating dependency on tribal knowledge and improving response accuracy for support teams.
- Additionally, engineered **scalable data pipelines and database migration workflows**, including an ETL system handling **10M+ records with zero data loss**, and implemented CI/CD-based schema versioning using Flyway and GitHub Actions to ensure reliable and automated deployments.

Projects

Open-Source Cloud Resource Governance Engine (CRGE) | Full-Stack DevOps & AI

Sept 2025 – Nov 2025

- Engineered a Policy-as-Code SaaS layer** with a Python backend and **AI Compliance Engine** to validate resource requests, reducing security risks and **optimizing resource consumption** before provisioning.
- Developed and managed the PaaS orchestration** using **Kubernetes (k3s)**, **PostgreSQL**, and **Celery/Redis**; implemented **HPA** and **Readiness Probes** to ensure service resilience and dynamic scaling under load.
- Authored reusable Terraform modules** and implemented **MinIO** for remote **IaC State Management** with locking, adhering to production-grade IaaS security and operational principles.
- Implemented a full SRE Observability stack (Prometheus/Grafana/Loki)** and a **CI/CD pipeline** for automated, zero-downtime deployment and real-time performance monitoring of all microservices.

Education

B.Tech in Information Technology, Jeppiaar Engineering College, Chennai

Aug 2019 – Jul 2023

- CGPA: 8.8/10

Certifications & Awards

- Microsoft Azure AI Engineer Associate | AWS Cloud Practitioner
- Google Text Prompt Engineering | Google Data Analytics Professional
- IBM AI Associate Data Scientist | IBM Enterprise Design Thinking – AI
- IBM Growth Excellence Award (2024), Exemplar Award (2025)
- Finalist, IBM Watsonx Global Challenge 2025